BUSINESS DOMAIN & SYSTEM OVERVIEW

The system is designed for a **Condominium Access Control & Dues Collection System**. It is a Java-based console prototype that provides a centralized way to manage tenant directories, monthly dues & late penalty calculations, security gate access logs, and visitor pass records.

TARGET USERS

Condo Administrator / Building Management – Manages tenant directory records, updates tenant billing information, and processes monthly dues and late penalty statements.

Security Gate Staff / Guards – Validates unit registration, processes resident and visitor gate entry, issues temporary visitor passes, and views security access logs.

PROBLEM STATEMENT

Condominium management offices and security personnel need to keep track of resident directories and entry logs while accurately calculating monthly dues and late payment penalties. Managing these operations manually can make it difficult to verify resident identities, track overdue payments, issue visitor passes, and maintain accurate entry logs. The proposed system organizes these activities by providing a centralized Java application for recording tenant data, calculating dues/penalties, and managing gate access logs.

CORE SYSTEM FEATURES

Tenant Directory Management – Registers new tenants, displays directory records, searches units, updates tenant details, and removes tenant records.

Calculate Dues & Late Penalties – Computes monthly association dues ($\text{sqm} \times 5.00$) and calculates flat or daily tiered late penalties based on payment overdue days.

Gate Access Control – Processes entry requests for residents, visitors, and service personnel with real-time timestamping (yyyy-MM-dd and hh:mm:ss a).

Visitor Pass Issuance – Verifies resident host availability and dynamically generates 4-digit visitor passes (Pass ID).

Security Access Logs – Maintains and displays dedicated entry logs for verified residents and visitors.

LOGIC FLOW

$$\text{Select Menu Option} \longrightarrow \text{Search / Verify Unit Registration} \longrightarrow \text{Validate Input Data} \longrightarrow \text{Process System Logic / Calculation} \longrightarrow \text{Update Memory Data / Generate Logs}$$
The system uses variables to store tenant details, unit dimensions, payment overdue days, and entry timestamps. Operators perform calculations (dues and penalty computations) and string comparisons, while decision conditions (if-else and switch statements) determine whether gate access is granted or if a penalty tier applies.

SYSTEM PROCESS SPECIFICATIONS

TITLE: Condominium Access Control & Dues Collection System

USER: Condo Administrator / Building Management and Security Gate Staff

PROCESSES

1.Main Menu Navigation – Routes users to Tenant Directory, Dues Calculation, Security Access, or System Exit.

2.Tenant Registration & Directory – Captures Unit Number, Tenant Name, Base Rent, and Unit Size (sqm) into system memory.

3.Monthly Dues & Penalty Calculation – Calculates association dues ($\text{sqm} \times 5.00$), evaluates overdue days, applies flat (1–5 days) or daily tiered (>5 days) penalties, and prints itemized billing statements.

4.Resident Entry Processing – Cross-checks entered unit and resident name against registered tenant records to grant or deny access.

5.Visitor Pass Management – Verifies host resident registration, captures host approval status, and issues a randomized 4-digit pass ID upon approval.

6.Service / Delivery Personnel Access – Issues temporary work permits linked to a target unit for service company workers.

7.Access Log Display – Formats and prints structured historical logs for resident entries and visitor passes.

SCENARIO

A resident or visitor arrives at the security gate, or a resident visits the building management office for billing. The security guard or administrator selects the appropriate module. If calculating dues, the system checks if the unit is registered to auto-populate details, accepts overdue days, calculates tiered penalties, and outputs an itemized statement. If processing gate entry, the system verifies unit registration and resident identity or host approval. If verified, access is granted, a timestamped log or visitor pass is generated, and the access record is saved in memory.